

Experiment No.	7
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models ¹
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Submission Date	12/08/26

1 Objective

To understand ensemble learning strategies – Bagging, Boosting, and Stacking – implement a Bagging classifier and Boosting classifiers (AdaBoost and Gradient Boosting), build a Stacked Ensemble using multiple heterogeneous base learners, compare all three in terms of accuracy, stability, and generalization, and analyze the effect of ensemble methods on bias and variance.

2 Problem Statement

Given 30 numerical features extracted from digitized images of breast mass fine-needle aspirates, classify each sample as Malignant or Benign using three different ensemble strategies, and compare their predictive performance.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer Dataset
Dataset Source	UCI Machine Learning Repository (also available via <code>sklearn.datasets.load_breast_cancer()</code>)
Number of Samples	569
Number of Features	30 numerical attributes
Number of Classes	2 (Malignant, Benign)
Missing Values	0
Train-Test Split	80% train (455 samples) / 20% test (114 samples), stratified, <code>random_state=42</code>

4 Exploratory Data Analysis

- Dataset overview and missing-value check (none found)
- Class distribution (357 Benign, 212 Malignant)
- Correlation heatmap across all 30 features
- Feature distribution histograms

Inference: the dataset is moderately imbalanced (about 63% Benign / 37% Malignant) but clean, with no missing values requiring imputation. Several features (e.g.

mean radius, mean perimeter, mean area) are strongly correlated with each other, as expected since they all describe cell size from different angles – this redundancy is exactly the kind of setting where ensemble methods (which combine many models on different feature/sample subsets) tend to perform well.

5 Data Preprocessing

- **Missing values:** none present, no imputation needed.
- **Feature scaling:** all 30 features were standardized using `StandardScaler`, fit on the training set only, since the Stacked Ensemble's SVM base learner is distance-based and sensitive to feature scale.
- **Train-test split:** 80/20, stratified on the target to preserve the Malignant/Benign ratio in both sets.

6 Machine Learning Algorithm

Bagging (Bootstrap Aggregation)

Working principle: trains multiple Decision Trees, each on a different bootstrap sample (random sample with replacement) of the training data, and aggregates their predictions by majority vote.

Advantages: reduces variance; trees are trained independently and in parallel; robust to overfitting compared to a single deep tree.

Limitations: does not reduce bias – if the base learner is systematically wrong, bagging alone won't fix it.

Boosting (AdaBoost / Gradient Boosting)

Working principle: trains models sequentially, with each new model focused on correcting the errors of the previous ones – AdaBoost reweights misclassified samples, Gradient Boosting fits each new model to the residual errors of the ensemble so far.

Advantages: reduces bias; often achieves higher accuracy than bagging on the same base learner.

Limitations: sequential training means it can't be parallelized as easily; more prone to overfitting on noisy data since it keeps focusing on hard examples.

Stacked Ensemble

Working principle: trains several heterogeneous base models (SVM, Naive Bayes, Decision Tree), then trains a meta-learner (Logistic Regression) on the base models' predictions to learn the optimal way to combine them.

Advantages: can outperform any individual base model by exploiting their diverse error patterns; flexible – any classifier can be a base model or meta-learner.

Limitations: more complex and computationally expensive to train; harder to interpret than a single model.

7 Experimental Setup

Python Version	3.11
Scikit-Learn Version	1.9.0
IDE	Jupyter Notebook
Operating System	Windows
Hardware	Intel i5, 16GB RAM

8 Hyperparameter Selection

Parameter	Search Space
Bagging <code>n_estimators</code>	10, 50, 100
Bagging <code>max_samples</code>	0.5, 0.7, 1.0
Bagging <code>max_features</code>	0.5, 0.7, 1.0
Boosting <code>n_estimators</code>	50, 100, 150
Boosting <code>learning_rate</code>	0.01, 0.1, 1.0
Gradient Boosting <code>max_depth</code>	2, 3, 4
Cross-Validation Folds	5
Search Method	GridSearchCV

9 Results

Table 1: Bagging Hyperparameter Evaluation (top configurations)

<code>n_estimators</code>	<code>max_samples</code>	<code>max_features</code>	Avg CV Accuracy (%)
100	1.0	0.5	96.04
100	0.7	0.5	95.82
50	1.0	0.5	95.82
50	1.0	0.7	95.82
100	1.0	0.7	95.82
50	0.5	0.5	95.60
100	0.5	0.7	95.60
50	0.7	0.7	95.38

Best: `n_estimators`=100, `max_samples`=1.0, `max_features`=0.5 (Avg CV Accuracy 96.04%, Avg CV F1 0.9684)

Table 2: Boosting Hyperparameter Evaluation

Model	<code>n_estimators</code>	<code>learning_rate</code>	<code>max_depth</code>	Avg CV Accuracy (%)	Avg CV F1
AdaBoost	100	1.0	–	97.36	0.9792

Gradient Boosting	100	1.0	2	97.58	0.9808
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Gradient Boosting was selected as the best boosting variant and carried forward to final evaluation.

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM, Naive Bayes, Decision Tree	Logistic Regression	96.70	0.9738

Table 4: Performance Comparison of Ensemble Models (Test Set)

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	94.74	0.9583	0.9583	0.9583
Boosting (Gradient Boosting)	96.49	0.9595	0.9861	0.9726
Stacked Ensemble	97.37	0.9726	0.9861	0.9793

AUC (test set): Bagging = 0.9926, Boosting (Gradient Boosting) = 0.9931, Stacked Ensemble = 0.9934.

10 Result Visualization

Include, from the notebook: class distribution, correlation heatmap, feature histograms, confusion matrices for all three final models, the overlaid ROC curve plot, and the train-vs-test accuracy bar chart.

Inference (ROC/AUC): all three models achieve AUC above 0.99, meaning each one separates malignant from benign cases very well across nearly all decision thresholds – the differences between them are concentrated in the small number of borderline cases near the default 0.5 threshold, which is exactly where the final Accuracy/Precision/Recall/F1 numbers in Table 4 differ.

Inference (Bias-Variance chart): Bagging and Boosting both reach 100% training accuracy but drop to 94.74% and 96.49% respectively on the test set – a noticeable train-test gap indicating some overfitting, more pronounced for Bagging. The Stacked Ensemble shows a smaller gap (98.46% train vs 97.37% test), suggesting its meta-learner is generalizing better rather than memorizing the training set.

11 Discussion

How does Bagging reduce variance?

By training many Decision Trees on different bootstrap samples and averaging their votes, the individual trees' overfitting mistakes (which differ from tree to tree since each sees different data) tend to cancel out in the aggregate, producing a smoother, less variance-prone decision boundary than any single deep tree.

How does Boosting address model bias?

Each new model in the boosting sequence is trained specifically to correct the errors of the previous ensemble (via reweighting in AdaBoost, or fitting to residuals in Gradient Boosting). This lets a sequence of otherwise weak/biased learners combine into a much stronger overall model, directly attacking underfitting rather than just averaging over variance.

Why does stacking benefit from heterogeneous models?

SVM, Naive Bayes, and Decision Tree make errors on different examples because they rely on different assumptions (margin maximization, feature independence, axis-aligned splits, respectively). A meta-learner trained on their combined outputs can learn which base model to trust more in which situation, capturing patterns no single base model sees alone – which is reflected here in the Stacked Ensemble outperforming both individual ensemble strategies on the test set.

Which ensemble method performed best and why?

The **Stacked Ensemble** performed best overall (97.37% test accuracy, $F1 = 0.9793$, $AUC = 0.9934$), narrowly ahead of Gradient Boosting and more clearly ahead of Bagging. This matches the bias-variance analysis above: Stacking combines Bagging-style diversity (heterogeneous base models) with Boosting-style correction (a meta-learner that weighs each base model's strengths), while showing the smallest train-test accuracy gap of the three – indicating the best balance between fitting the data and generalizing to unseen samples.

12 Conclusion

All three ensemble strategies substantially outperform what a single base classifier would typically achieve on this dataset, confirming the core premise of ensemble learning. Bagging reduced variance at the cost of some residual overfitting; Boosting (specifically Gradient Boosting) reduced bias and achieved higher test accuracy than Bagging; and the Stacked Ensemble, by combining diverse base learners under a meta-learner, achieved the best overall test performance and the most balanced bias-variance trade-off of the three.

13 References

1. Scikit-learn: Ensemble Methods, <https://scikit-learn.org/stable/modules/ensemble.html>
2. Scikit-learn: Bagging Classifier, <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html>
3. UCI Machine Learning Repository: Breast Cancer Wisconsin (Diagnostic) Dataset